



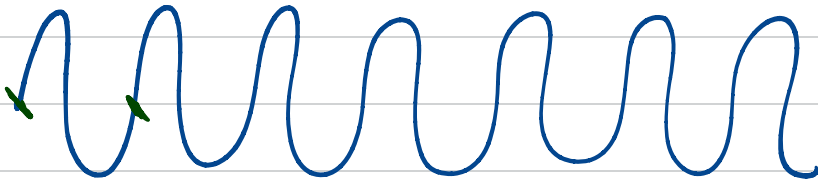
LoRA = Rules + Terminologies + Parlance

Frequency
Sampling Rate
Amplitude
Mel Spectrogram.

90°
↓
APL



Sound → Vibration.



one vibration / cycle → Frequency

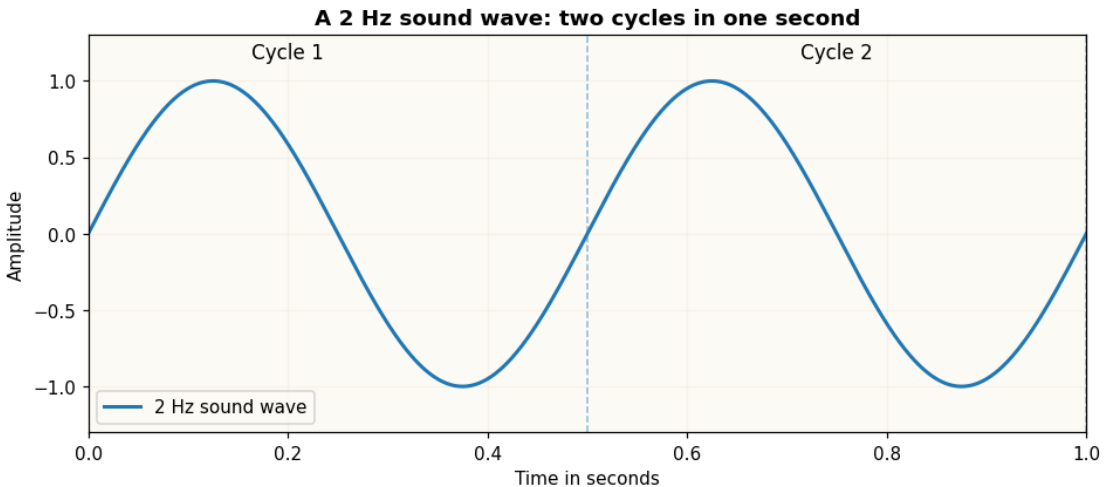
Audio has 100Hz -> vibrating 100 times in a second.

15000 Hz $\rightarrow$ Vibrate at this freq.

Vowels $\rightarrow$ low freq.

Consonants $\rightarrow$ high freq.

Thumping sound $\rightarrow$ low freq.



how quickly do you want to poll it.

Nyquist Frequency: your sampling rate should at least be double for frequency.



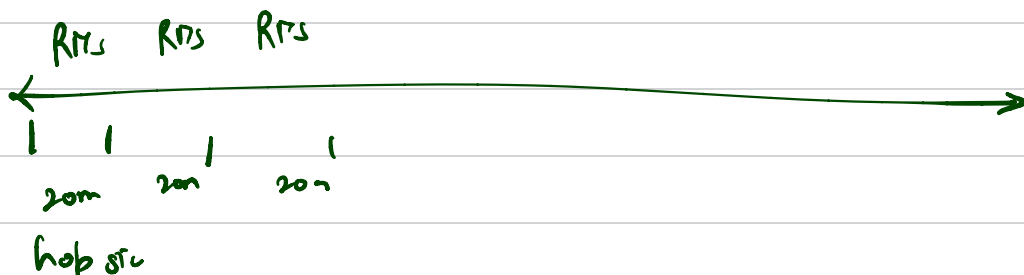
source : LibriSpeech dummy
 samples : 93,680 sr: 16000 Hz
 dtype : float32 range: [-0.3
 reference: 'MISTER QUILTER IS THE

$$\frac{93680}{16000}$$

20 oronp

5 oron →

$$20/5 = 4 \text{ boxes}$$

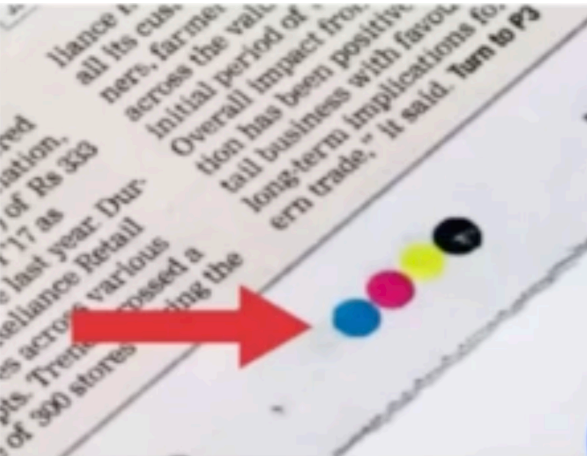


0: 1024 → chunk → RMS
256: 1280 → chunk → RMS

```
frames = librosa.util.frame(man, frame_length=1024, hop_length=256)
rms = np.sqrt((frames ** 2).mean(axis=0))
zcr = (np.abs(np.diff(np.sign(frames), axis=0)) > 0).mean(axis=0)
```

```
speech    = rms > rms.max() * 0.10           # loud enough to
voiced    = speech & (zcr < np.median(zcr[speech])) # periodic → vo
unvoiced  = speech & (zcr >= np.median(zcr[speech])) # noisy    → s
```

Have a 200 MP camera, can the image contain just 4 shades of colour?



8 bits

256 bits $\rightarrow$ Colour
Image

16 bits $\rightarrow$ MRI

